

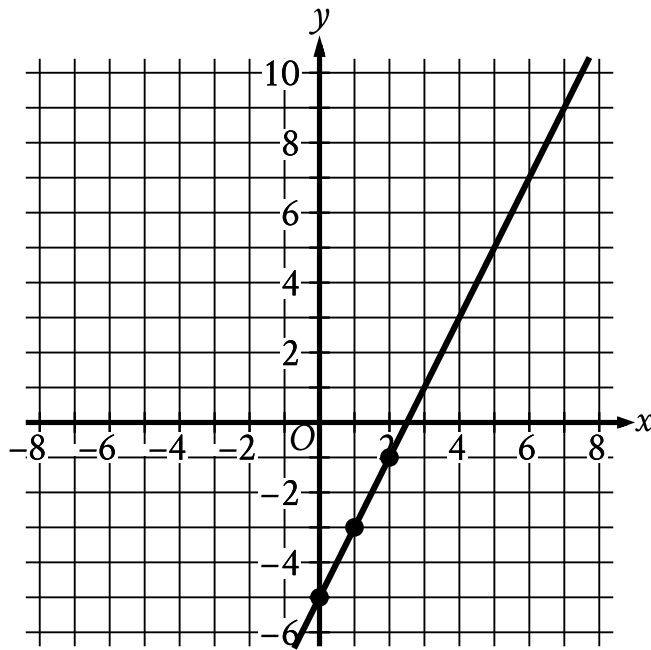
☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

19



- The line slants sharply up from left to right.
- The line passes through the following points:
 - (0 comma negative 5)
 - (1 comma negative 3)
 - (2 comma negative 1)

The graph shows the linear relationship between x and y . Which table gives three values of x and their corresponding values of y for this relationship?

A.

x	y
0	0
1	-7
2	-9

B.

x	y
0	0
1	-3
2	-1

C.

x	y
0	-5
1	-7
2	-9

D.

x	y
0	-5
1	-3
2	-1

Q2

ALGEBRA

What value of x is the solution to the equation $16x + 24 = 24x$?

A. -4

B. $\frac{3}{10}$

C. $\frac{1}{3}$

D. 3

Q3

ALGEBRA

The length, y , of a white whale was 162 centimeters cm when it was born and increased an average of 4.8 cm per month for the first 12 months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in cm, of the whale?

A. $y = 162x$

B. $y = 162x + 162$

C. $y = 4.8x + 4.8$

D. $y = 4.8x + 162$

Q4

ALGEBRA

Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the xy -plane and passes through the point $(0, 5)$. Which equation defines line j ?

- A. $y = 6x + 5$
- B. $y = -5x + 5$
- C. $y = -6x + 5$
- D. $y = 5x + 5$

Q5

ALGEBRA

$$w + 7 = 357$$

What value of w is the solution to the given equation?

- A. 51
- B. 350
- C. 364
- D. 3,577

$$f(x) = 4x + b$$

For the linear function f , b is a constant and $f(7) = 28$. What is the value of b ?

- A. 0
- B. 1
- C. 4
- D. 7

$$7x - 4y = -84$$

For the given equation, which table gives three values of x and their corresponding values of y ?

A.

x	0	4	8
y	21	28	35

B.

x	0	4	8
y	35	28	21

C.

x	21	28	35
y	0	4	8

D.

x	21	28	35
y	8	4	0

Q8

ALGEBRA

A librarian has 43 books to distribute to a group of children. If he gives each child 2 books, he will have 7 books left over. How many children are in the group?

- A. 15
- B. 18
- C. 25
- D. 29

Q9

ALGEBRA

The function g is defined as $g(x) = 5x + a$, where a is a constant. If $g(4) = 31$, what is the value of a ?

- A. 30
- B. 22
- C. 11
- D. -23

A store sells two different-sized containers of a certain Greek yogurt. The store's sales of this Greek yogurt totaled 1,277.94 dollars last month. The equation $5.48x + 7.30y = 1,277.94$ represents this situation, where x is the number of smaller containers sold and y is the number of larger containers sold. According to the equation, which of the following represents the price, in dollars, of each smaller container?

- A. 5.48
- B. $7.30y$
- C. 7.30
- D. $5.48x$